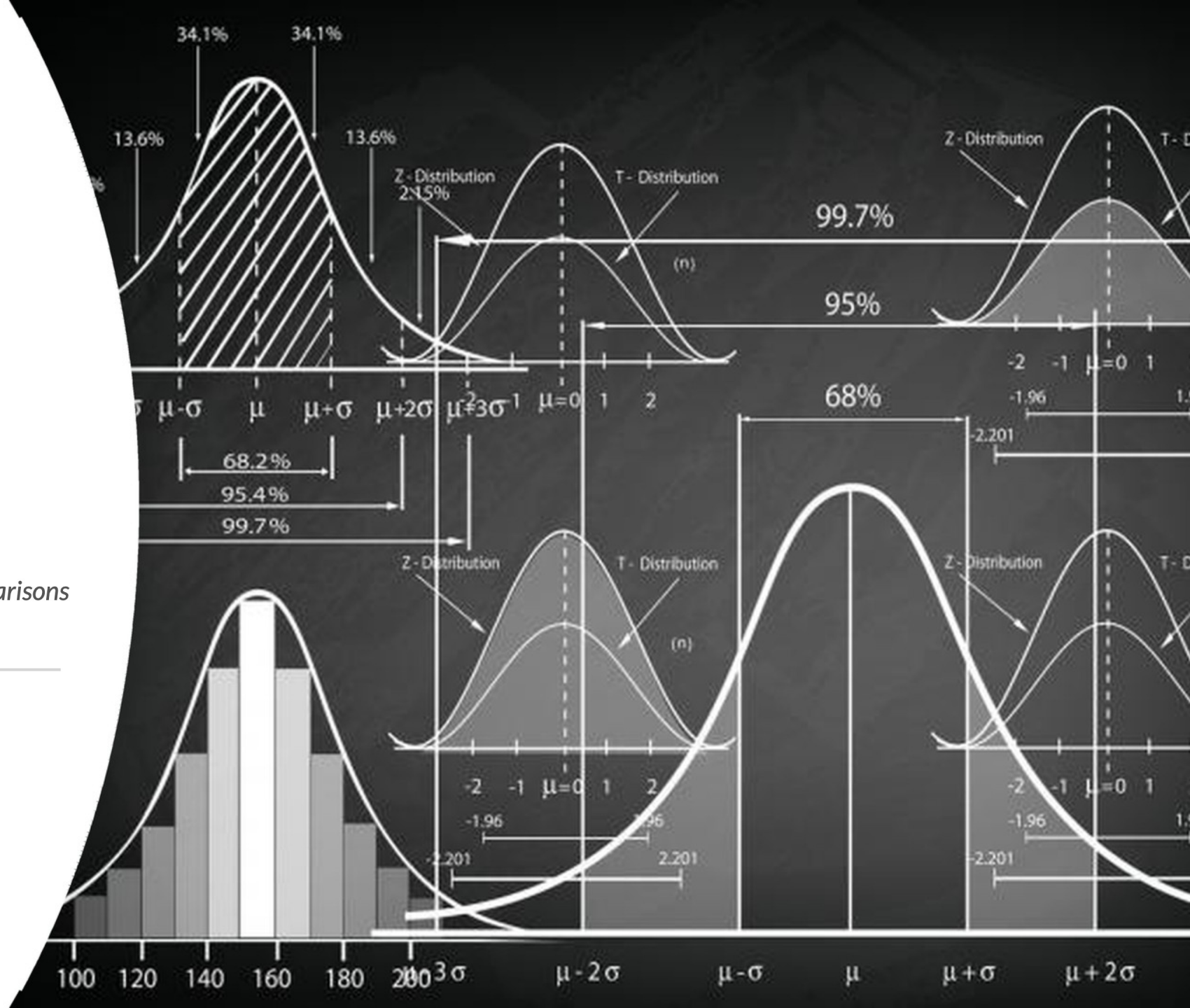
Comparing Mutual Information in Finite Samples

Can MI-specific degrees of freedom improve comparisons between populations?

By Daniel Zhao (*Master of Computer Science*)

Supervised by Assoc. Prof. Joseph Lizier



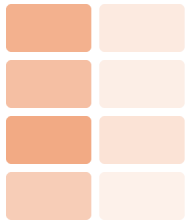
Motivation: Comparing MI Across Populations

Mutual information quantifies how much knowing one variable reduces uncertainty about another. Two different estimates do not prove different population MI.

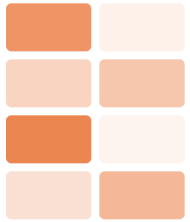
Null hypothesis $H_0 : \; I(P) = I(Q)$ *the tables may differ; only their MI must match*

Example: Online Shoppers

New visitors
1,694 sessions

→  → **Î(P) = 0.0043**

Returning visitors
10,551 sessions

→  → **Î(Q) = 0.0179**

Month × purchase tables; corrected MI estimates in nats.

The Statistical Task

Draw a count table from each population

↓

Estimate MI from each table

↓

Real difference, or sampling noise?

Literature Review: What Exists and the Gap

Existing work supplies the MI estimator, a Normal Wald comparison and the Welch–Satterthwaite principle. It does not supply MI-specific degrees of freedom.

	A *MI Estimation and Uncertainty*	B *Normal Wald Comparisons*	C *Welch–Satterthwaite Moment Matching*
Idea	Plug-in MI with a leading bias correction; first-order variance V(P)/n	Divide the MI difference by its standard error; compare with N(0, 1)	Match an estimated variance to a scaled χ^2 to set Student t degrees of freedom
Key work	Miller 1955; Moddemeijer 1989; Brillinger 2004; Paninski 2003	Wald 1943; Mora and Ruiz-Castillo 2009	Satterthwaite 1946; Welch 1947; Hutcheson 1970; Pan and Wall 2002
Gap	Gives the variance of MI, but not how much its variance estimate varies	Same normal cutoff at every sample size, however noisy the standard error	For MI, needs the sampling variance of the pointwise-MI variance estimate

This thesis derives that sampling variance, giving MI-specific degrees of freedom.

Method: One Statistic, Two Distributions

Both tests compute the same statistic from the same two tables. Expanded Welch changes only the distribution used to judge it.

Shared statistic

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}} \qquad \widehat{\mathrm{SE}}^{\,2} = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}$$

Pointwise-MI variance ÷ sample size estimates each MI estimate's sampling variance.

Normal Wald

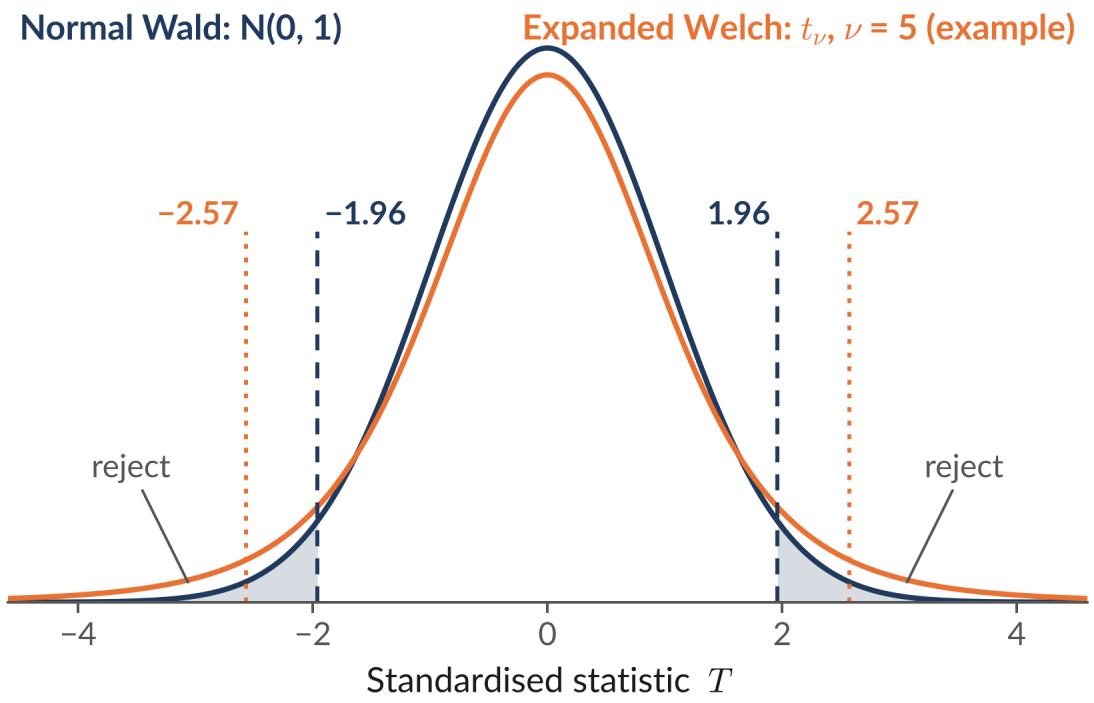
Compare T with N(0, 1). Reject when |T| > 1.96 at every sample size.

$$p_{\mathrm{Wald}} = 2\{1 - \Phi(|T|)\}$$

Expanded Welch

Compare T with a Student t. Fewer degrees of freedom give a larger cutoff.

$$p_{\mathrm{EW}} = 2\{1 - F_t(|T|;\, \widehat{\nu})\}$$



Approach & Contribution

I derive how the estimated pointwise-MI variance changes with the table, which gives MI-specific degrees of freedom for a Student t test.

Derivation Chain

Cell-probability change

One observation moves probability toward cell (x, y)

→

Margins and pointwise MI change

Row and column totals shift, so every log ratio shifts

→

Variance sensitivity (new)

g(x, y): first-order change in the pointwise-MI variance

→

Degrees of freedom (new)

The spread of these sensitivities sets the Student t

Result and Approximations

$$\widehat{\nu}_P = \frac{2\, n_P\, \widehat{V}(P)^2}{\widehat{\tau}^2(P)}$$

$\tau^2(P)$: how much each observation's effect on the variance estimate varies across the table.

More variability → fewer degrees of freedom → larger cutoff.

Two approximations, tested in the evaluation

1. First-order expansion of the variance estimate
2. Scaled χ^2 moment matching (Satterthwaite)

Evaluation Design

Simulations from populations with known MI answer three research questions. Both tests are applied to the same sampled tables.

Three Research Questions

1. **Approximation accuracy**
   Do the approximations predict the mean and sampling variance of the variance estimate, and the degrees of freedom?

2. **False positives**
   Under equal population MI, how close is each test's rejection rate to 5%?

3. **Power**
   When population MI differs, how often does each test reject, before and after matching false-positive rates?

Simulation Grid

1. **Known population MI**
   Copying construction fixes the margins; solve for each target MI

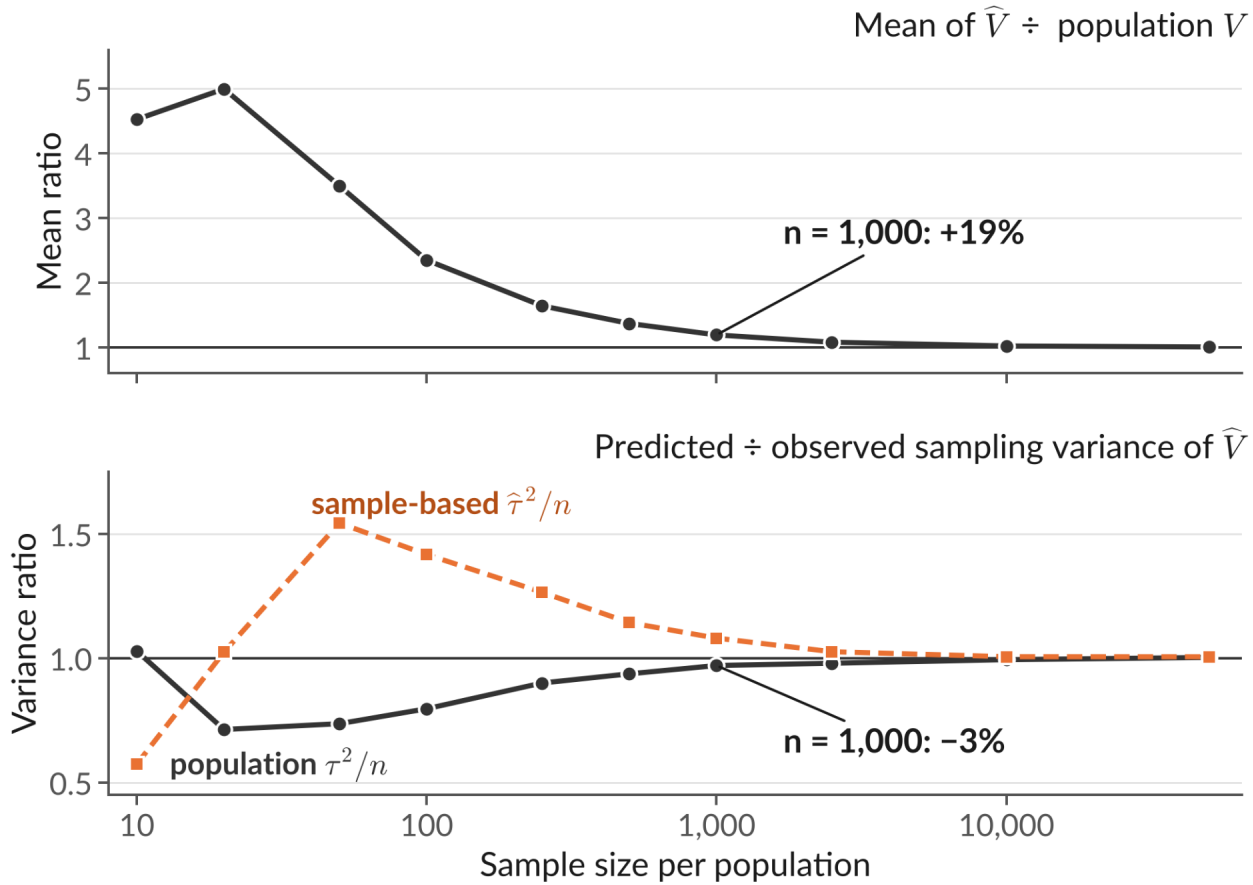
2. **Table size**
   2×2, 4×4, 8×8 and 16×16 (4 to 256 cells)

3. **Margin regime**
   Uniform, same skew and different skew

4. **100,000 repetitions per setting**
   Fitting and evaluation samples drawn independently

Results: RQ1 Approximation Accuracy

The variance approximations improve as sampling becomes adequate, but the mean can stay inflated while the variance is predicted well.



Different-skew P · 4×4 · MI 0.02 nats

Mean: rare cells inflate it

A cell seen once gets an extreme pointwise MI. Its squared deviation pushes the variance estimate up until counts repeat.

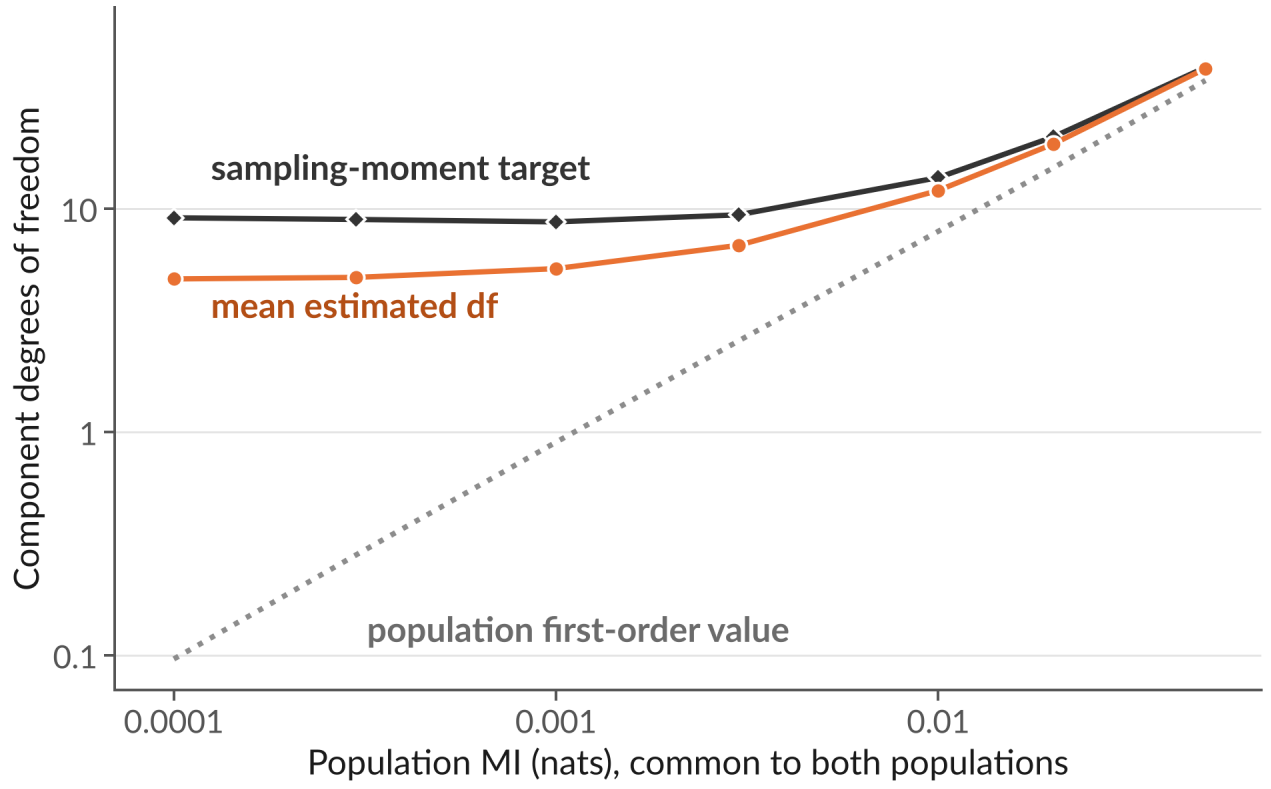
Variance: lags, then agrees

Rare cells appear in some samples and not others. The first-order prediction misses these jumps at moderate n.

At n = 1,000: variance prediction 3% low, mean 19% high. Both enter the degrees of freedom.

Results: RQ1 Near Independence

Near independence, sampling noise becomes central: estimated degrees of freedom stay near 5 while the first-order value collapses.



4×4 · n = 1,000 fixed · population P

First-order value collapses

At independence every pointwise MI is zero, so the linear sampling term vanishes. The first-order value is about n × I(P).

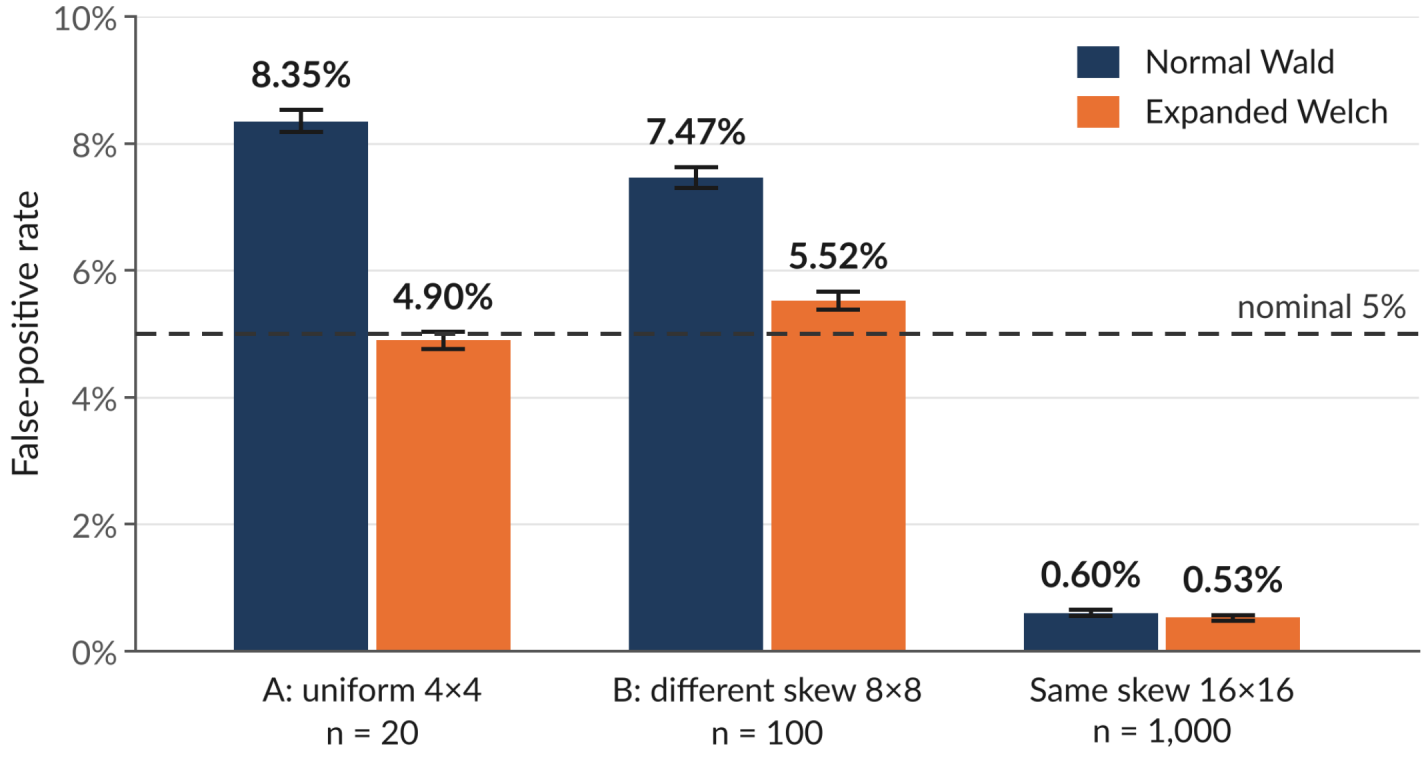
Estimated df stay near 5

Quadratic sampling errors give each table apparent association. The estimate tracks that noise: about G / 2, with G = 2n × Î(P).

At MI = 0.0001: mean estimated df 4.9, target 9.1, first-order 0.1

Results: RQ2 False Positives

A larger cutoff helps where Wald over-rejects, but deepens conservatism where Wald is already below 5%.



Equal population MI of 0.02 nats, so every rejection is a false positive. Whiskers: 95% intervals.

✓ **Wald over-rejects**

29 of 38

grid settings move closer to 5%

⚠ **Wald already below 5%**

0 of 82

can improve; 48 move farther away

Results: RQ2 Bias in the MI Difference

Changing the cutoff cannot repair a biased MI difference: similar variance errors give opposite false-positive behaviour.

16×16, n = 1,000, equal MI	Variance ratio	Bias ÷ SD	Normal Wald	Exp. Welch
Same skew (identical tables)	1.93	≈ 0	0.60%	0.53%
Different skew	1.94	2.06	26.01%	24.84%

Variance ratio: average squared SE ÷ observed variance of the MI difference. Rates are false-positive rates.

Same skew: SE too large

The squared SE is about 1.9 times the true variance. |T| shrinks, so both tests reject far too rarely.

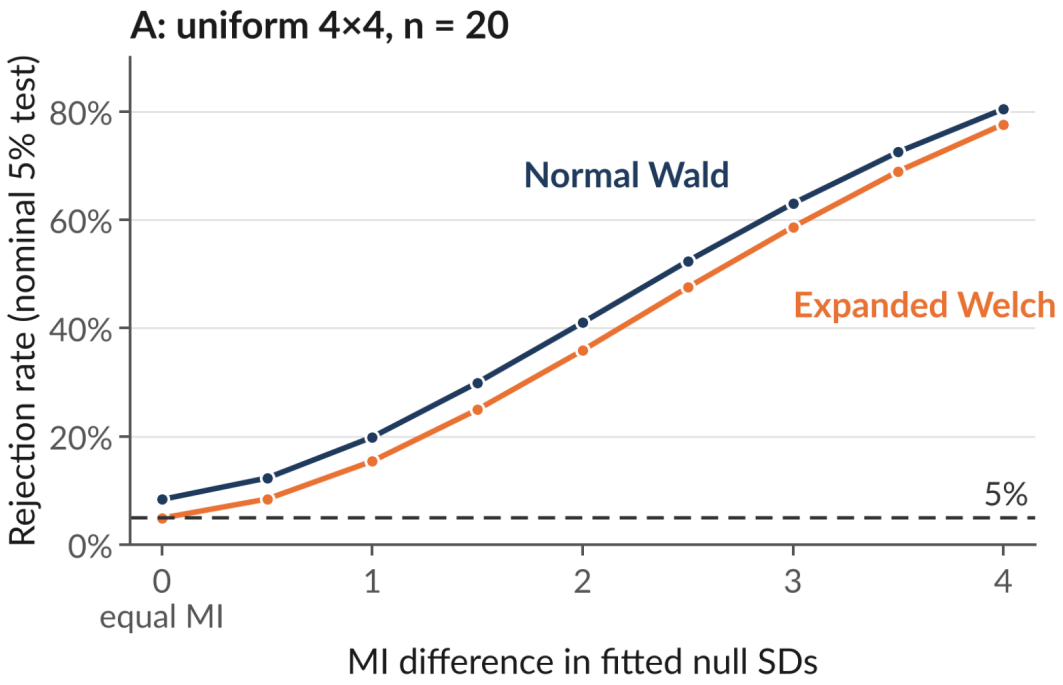
Different skew: SE too large, plus bias

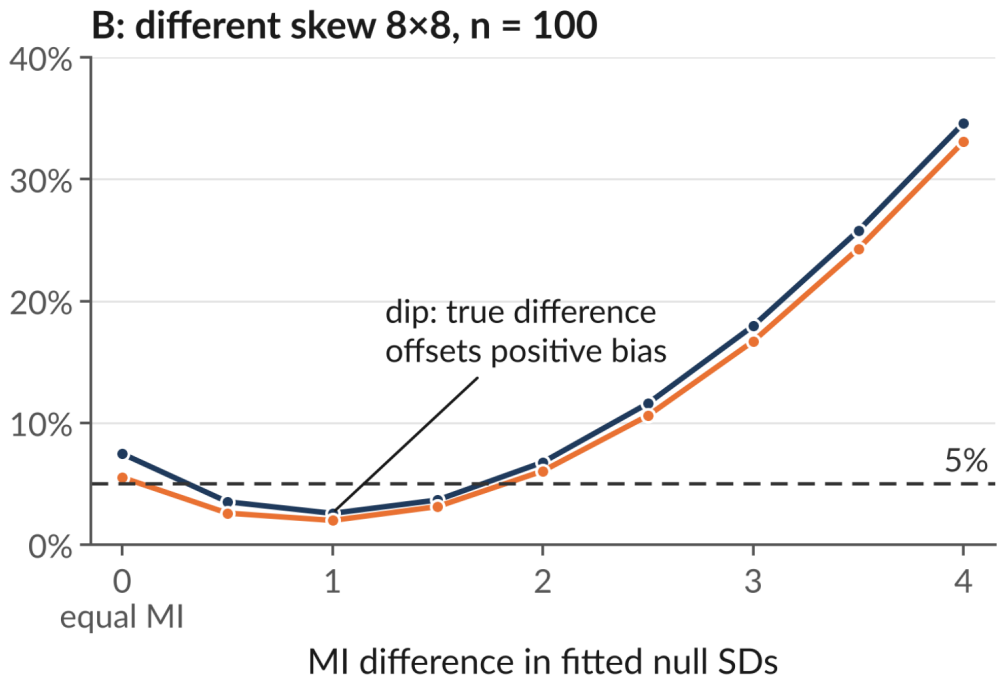
The same variance error, plus a bias of +0.030 nats (2.06 SDs). The estimate sits away from zero, so both tests reject far too often.

Expanded Welch changes the cutoff but keeps the shifted estimate: 26.01% → 24.84%

Results: RQ3 Nominal Power

Expanded Welch's lower false-positive rate comes with lower nominal power, so a fair comparison must match false-positive rates first.





A: Expanded Welch sits below Wald at every difference.

B: both dip first, as the true difference offsets a positive bias.

Horizontal axis: population MI difference in fitted null SDs; zero is the equal-MI case. Vertical scales differ between panels.

Results: RQ3 Calibrated Power

After matching false positives, Expanded Welch can gain power, but the gain depends on the equal-MI setting used for calibration.

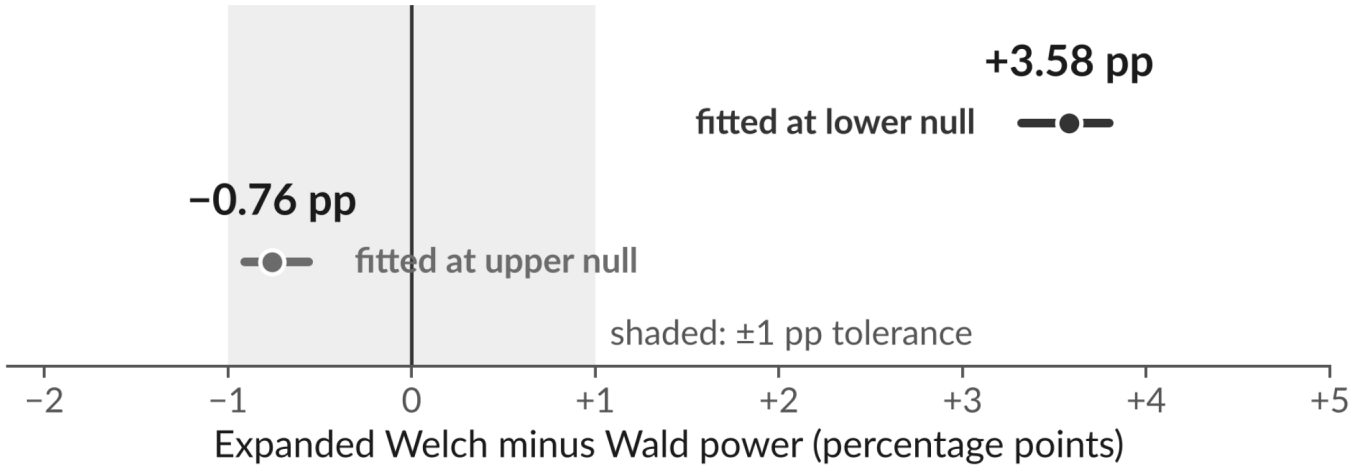
1 Fit
Pick each threshold for 5% rejection under equal MI

→

2 Hold fixed
Keep both thresholds fixed for every alternative

→

3 Evaluate
Apply them to independent null and alternative samples



Setting A, 2 null SDs

Alternative: MI 0.02 vs 0.33. Lower null: both 0.02. Upper null: both 0.33.

Same alternative samples. Only the thresholds change, and the gain reverses.

Illustration: Online Shopping Data

Returning visitors show higher month–purchase MI, and both tests reject equal MI. These data cannot show which test is better calibrated.

Month × Purchase · corrected MI (nats)

New visitors (n = 1,694)

0.0043

Returning visitors (n = 10,551)

0.0179

Difference −0.0136 nats, SE 0.0033, so T = −4.07

Normal Wald: $p = 4.7 \times 10^{-5}$ · Holm 0.00019

Expanded Welch (26.8 df): $p = 3.7 \times 10^{-4}$ · Holm 0.0015

✓ **Both reject equal MI after Holm adjustment**

Weekend × Purchase

New vs returning (nats)

0.00066 vs 0.00073

T = −0.06

p = 0.955 (Wald), 0.960 (Expanded Welch, 2.2 df)

Neither rejects

Population MI is unknown here, so this shows how to read the outputs. The simulations carry the evidence about reliability.

Data: Online Shoppers Purchasing Intention (Sakar and Kastro, 2018). Holm adjustment over a family of six comparisons.

Conclusion & Next Steps

Expanded Welch improves calibration in particular tested settings. Estimation error and calibration dependence limit a broader recommendation.

Contribution & Findings

1

MI-specific variance sensitivity derived

Tracks cell weights, margins and pointwise MI to give MI-specific degrees of freedom

2

Selected false-positive improvements

Closer to 5% at A, B and 29 of 38 over-rejecting settings; more conservative elsewhere

3

Power gains depend on calibration

Gains after matching false positives can reverse when the fitting null changes

Next Steps

1. **Better bias and SE estimates** for sparse tables, each tested on its own
2. **Higher-order corrections**, starting with the second-order mean of the variance estimate

Appendix: Backup Slides

Supporting material for the question period. Each slide answers one likely question.

B1 **Variance-sensitivity derivation**
What exactly is new in the derivation?

B2 **Moment matching**
Why does this give those degrees of freedom?

B3 **Approximation ratios, all tables**
Is the example shown representative?

B4 **Degrees-of-freedom ratios, all tables**
How do moment errors move the df?

B5 **Near-independence calculation**
Why do estimated df stay positive?

B6 **Simulation and validity**
Could invalid outputs or noise explain this?

B7 **Robustness checks**
Does it persist under another construction?

B8 **Calibration validation**
Does calibration give a generally valid 5% test?

B9 **Application details**
How were the examples selected and adjusted?

B10 **References**
Full citations for the works named in the talk

Appendix: Variance-Sensitivity Derivation

The derivation tracks both the probability weights and the pointwise-MI values as one observation changes the table.

$$P_\varepsilon = (1-\varepsilon)P + \varepsilon\,\delta_{(x,y)}, \qquad g_P(x,y) = \left.\frac{d\,V(P_\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0}$$

$$g_P(x,y) = \underbrace{\{\ell_P(x,y) - I(P)\}^2 - V(P)}_{\text{probability weights change}} + \underbrace{2\{\ell_P(x,y) - \mathbb{E}_P[\ell_P \mid X{=}x] - \mathbb{E}_P[\ell_P \mid Y{=}y] + I(P)\}}_{\text{pointwise-MI values change with the margins}}$$

The first term holds pointwise-MI values fixed. The second tracks how changed row and column totals move every log ratio.

$$\widehat{V}(P) - V(P) \approx \frac{1}{n_P}\sum_{a=1}^{n_P} g_P(Z_a) \quad\Longrightarrow\quad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P}, \qquad \tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}$$

Existing: MI's influence function and its variance V(P)/n. **New here:** the sensitivity of V(P) itself, g, and its variance $\tau^2(P)$.

Thesis Section 4.2, Equation 4.30.

Appendix: Moment Matching

Satterthwaite's principle matches the mean and variance of the variance estimate to a scaled χ^2, for each sample and then combined.

Scaled χ^2 model for one component

$$\widehat{V}(P) \approx \mathbb{E}\{\widehat{V}(P)\}\,\frac{\chi^2_\nu}{\nu} \quad \Longrightarrow \quad \nu_{P,\mathrm{mom}} = \frac{2\,[\mathbb{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}$$

First-order value and table estimate

$$\nu_{P,\mathrm{FO}} = \frac{2\,n_P\,V(P)^2}{\tau^2(P)} \qquad \widehat{\nu}_P = \frac{2\,n_P\,\widehat{V}(P)^2}{\widehat{\tau}^2(P)}$$

Combine the two samples

$$\widehat{\nu}_{\mathrm{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\widehat{\nu}_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\widehat{\nu}_Q}$$

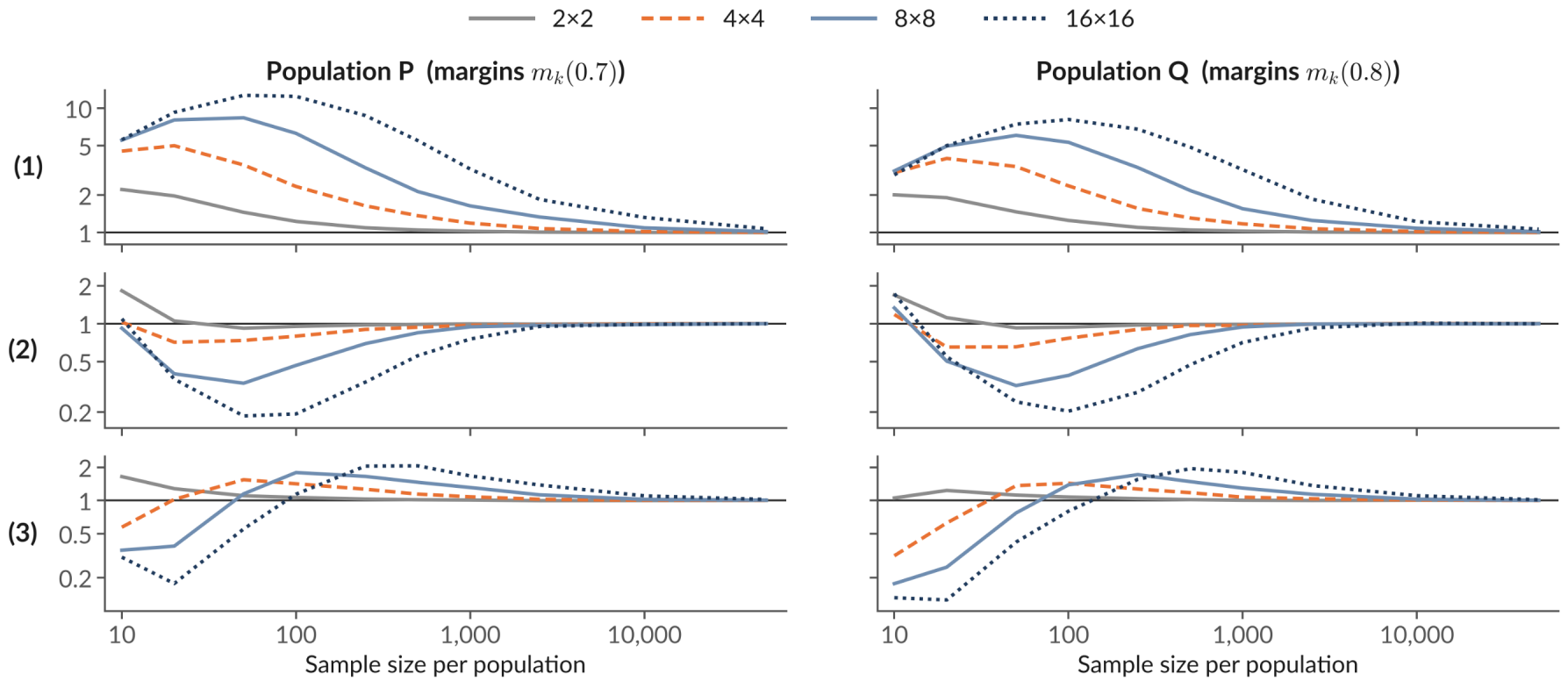
Dividing by n leaves the matched degrees of freedom unchanged. The larger variance contribution dominates the combination.

Worked 3×3 example: component df 16.65 and 45.86, combined 59.03. Wald p = 0.147, Expanded Welch p = 0.153.

Thesis Sections 4.1, 4.3 and 4.4; Satterthwaite (1946); Welch (1947).

Appendix: Approximation Ratios

Larger tables show higher mean peaks, deeper variance troughs and slower convergence, but every ratio approaches one.

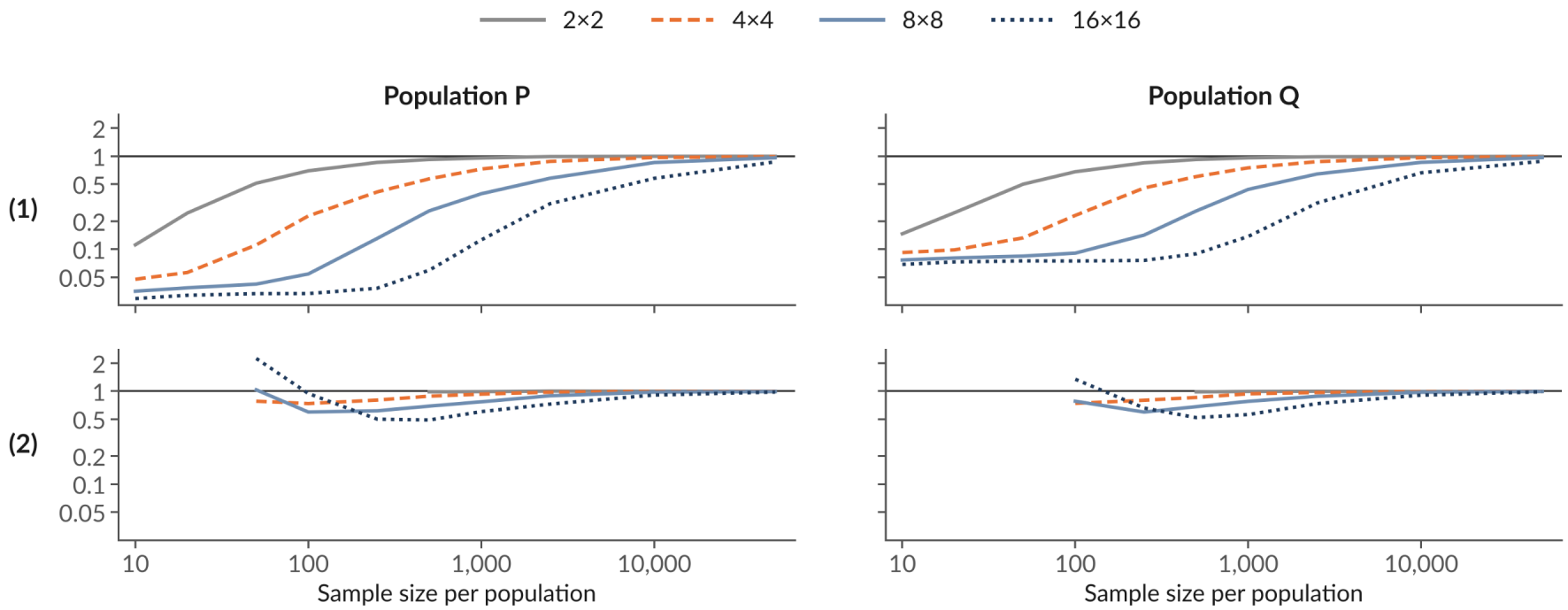


(1) mean of the variance estimate ÷ V · (2) population τ^2/n ÷ observed variance · (3) sample-based τ^2/n ÷ observed variance

Different-skew margins, MI 0.02 nats, 100,000 repetitions; one means agreement. Redrawn from thesis Figure 6.1.

Appendix: Degrees-of-Freedom Ratios

A mean that runs high pulls the first-order degrees of freedom below the target, even when the variance prediction is accurate.



(1) population first-order df ÷ sampling-moment target · (2) mean estimated df ÷ target, where every estimate is valid

4×4 P at n = 1,000: mean 19% high, variance 3% low, so first-order df 15.30 vs target 21.03

Different-skew margins, MI 0.02 nats. Redrawn from thesis Figure 6.2.

Appendix: Near-Independence Calculation

Quadratic sampling errors give each table apparent association, so estimated df track G/2 while the first-order value tends to zero.

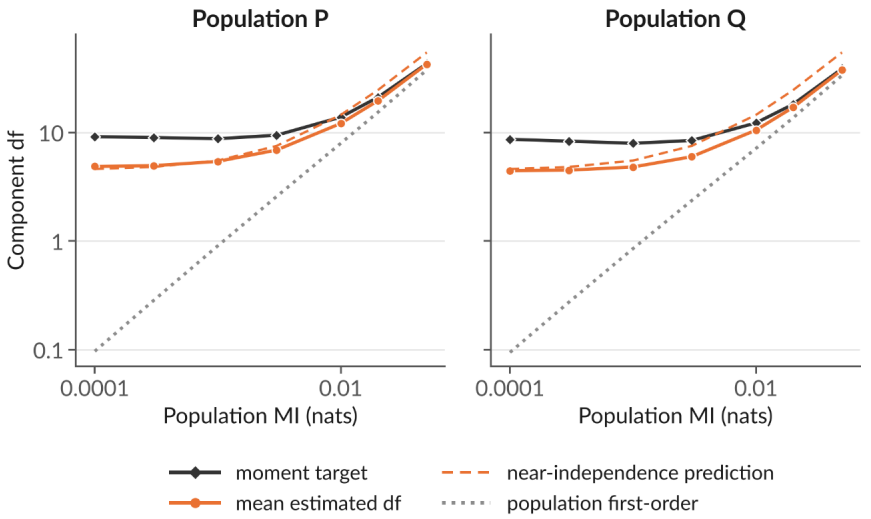
Population side: fixed-margin expansion

$$V(P) \approx 2\,I(P), \quad \tau^2(P) \approx 4\,V(P) \quad \Longrightarrow \quad \nu_{P,\mathrm{FO}} \approx n_P\,I(P)$$

Sample side: quadratic sampling errors (G statistic)

$$G = 2n\,\widehat{I}(P) \,\dot{\sim}\, \chi^2_d(\kappa), \qquad d = (r-1)(c-1), \qquad \kappa \approx 2n\,I(P)$$

$$\widehat{V} \approx \frac{G}{n}, \quad \widehat{\tau}^{\,2} \approx \frac{4G}{n} \Longrightarrow \widehat{\nu}_P \approx \frac{G}{2} \qquad \nu_{P,\mathrm{mom}} \approx \frac{(d+\kappa)^2}{d+2\kappa}$$



4×4, n = 1,000, MI 0.0001: d = 9 and κ ≈ 0.2, so mean estimated df ≈ 4.6 against a first-order value of 0.1. The plug-in variance prediction is about 1.96 times the actual sampling variance.

Thesis Appendix A and Figure 6.3. Population and sample-side predictions shown for the 4×4, n = 1,000 sweep.

Appendix: Simulation Design and Validity

Known populations, 100,000 repetitions per setting and explicit validity rules separate method behaviour from simulation noise.

Populations and Grids

$$p_{ij}(\lambda) = (1-\lambda)\, a_i a_j + \lambda\, a_i\, \mathbf{1}\{i=j\}, \qquad m_k(\rho) = \left(\rho,\ \tfrac{1-\rho}{k-1},\ \ldots,\ \tfrac{1-\rho}{k-1}\right)$$

Copy X into Y with probability λ, else draw Y independently. Margins stay fixed; λ is solved for the target MI.

Broad null grid: 4 table sizes × 10 sample sizes (10 to 50,000) × 3 margin regimes = 120 settings at MI 0.02.

Near independence: 4×4, n of 100, 1,000 and 10,000, MI from 0.0001 to 0.05.

Power: A, B and C, alternatives at 0 to 4 fitted null SDs.

Sampling, Validity and Uncertainty

Sampling: 100,000 multinomial repetitions per setting. PCG64 with seed 20261021 and separate streams for fitting and evaluation.

Validity: invalid outputs count as non-rejections. Minimum valid fraction on the grid: 0.958 for Wald, 0.631 for Expanded Welch. Both are valid in every repetition at A, B and C.

Uncertainty: Wilson 95% intervals for rates, paired intervals for method differences, and 1,000 joint bootstrap resamples for calibrated power.

Thesis Chapter 5 and Appendix B; design follows Morris, White and Crowther (2019).

Appendix: Robustness Checks

The improvement at A and B is specific to MI 0.02, while the sparse different-skew failure persists under another construction.

Check	Equal-MI setting	Normal Wald	Exp. Welch
Stronger common MI	A, common MI 0.33 nats	16.16%	13.56%
Stronger common MI	B, common MI 0.28 nats	7.62%	7.25%
Four-cell construction	16×16 different skew, n = 1,000	near 30%	near 30%
Larger sample	16×16 same skew, n = 10,000	4.38%	4.26%
Larger sample	16×16 different skew, n = 10,000	4.78%	4.68%

Stronger MI: both tests over-reject at A and B, even with the larger cutoff.

Four-cell construction: the sparse different-skew failure remains.

Larger samples: both 16×16 regimes return close to 5% at n = 10,000.

False-positive rates from thesis Tables 6.3 and 6.4 and Section 6.2.2.

Appendix: Calibration Validation

Thresholds fitted at one equal-MI setting do not control false positives at the other, so calibrated power gains are conditional.

Null rejection at A, 2 null SDs

Fitted at	Evaluated at	Wald	Exp. Welch
Lower null	Lower null	4.93%	4.99%
Lower null	Upper null	11.20%	13.71%
Upper null	Upper null	4.87%	4.84%
Upper null	Lower null	1.63%	0.79%

Lower null: both MI 0.02 nats. Upper null: both MI 0.3299 nats.

Each calibration hits about 5% only at its own fitting null.

Expanded Welch minus Wald power (pp)

Setting, h	Upper-null MI	Lower-null fit	Upper-null fit
A, 2	0.3299	+3.58	−0.76
A, 4	0.6399	+4.01	+0.18
B, 2	0.1503	+1.11	+0.05
B, 4	0.2806	+3.92	−0.39
C, 2	0.0263	+0.04	−0.06
C, 4	0.0327	+0.02	−0.01

Positive values favour Expanded Welch. Tolerance ±1 pp.

Thesis Tables 5.4, 6.5 and 6.6.

Appendix: Application Details

The shopping examples come from a family of six comparisons, so each method's p-values are Holm-adjusted.

Data: 12,330 sessions. Kept 1,694 New_Visitor and 10,551 Returning_Visitor sessions; excluded 85 Other.

Tables: month is 10 × 2 (no January or April sessions); weekend is 2 × 2. Zero cells kept, no smoothing.

Bias correction: 9/(2n) for month and 1/(2n) for weekend, per group.

Holm: order each method's six p-values, multiply by 6, 5, 4, 3, 2, 1, take the running maximum and cap at 1.

Month: difference −0.013563 nats, SE 0.003333, T = −4.069. Adjusted p 0.000189 (Wald), 0.00149 (Expanded Welch).

Six-comparison family · unadjusted p

Dataset	Predictor	Wald	Exp. Welch
Adult	Occupation	1.2×10^{-44}	1.6×10^{-42}
Adult	Education	1.9×10^{-34}	4.9×10^{-33}
Online Shoppers	Month	4.7×10^{-5}	3.7×10^{-4}
Student Perf.	Study time	0.198	0.234
Student Perf.	Mother's educ.	0.836	0.847
Online Shoppers	Weekend	0.955	0.960

Adult: female vs male records. Student Performance: schools GP vs MS.

Becker and Kohavi (1996); Cortez (2008); Sakar and Kastro (2018); Holm (1979).

Appendix: References

Key references underpinning the literature review, method and evaluation in this presentation.

[1] Bossomaier, T., Barnett, L., Harré, M., & Lizier, J. T. (2016). *An Introduction to Transfer Entropy*. Springer.

[2] Brillinger, D. R. (2004). Some data analyses using mutual information. *Brazilian J. Probability and Statistics*, *18*(2).

[3] Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian J. Statistics*, *6*(2).

[4] Hutcheson, K. (1970). A test for comparing diversities based on the Shannon formula. *J. Theoretical Biology*, *29*(1).

[5] Miller, G. A. (1955). Note on the bias of information estimates. In *Information Theory in Psychology*.

[6] Moddemeijer, R. (1989). On estimation of entropy and mutual information of continuous distributions. *Signal Processing*, *16*(3).

[7] Mora, R., & Ruiz-Castillo, J. (2009). The statistical properties of the mutual information index of multigroup segregation. UC3M Working Paper 09-84.

[8] Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, *38*(11).

[9] Pan, W., & Wall, M. M. (2002). Small-sample adjustments in using the sandwich variance estimator in GEE. *Statistics in Medicine*, *21*(10).

[10] Paninski, L. (2003). Estimation of entropy and mutual information. *Neural Computation*, *15*(6).

[11] Sakar, C., & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset. UCI Machine Learning Repository.

[12] Satterthwaite, F. E. (1946). An approximate distribution of estimates of variance components. *Biometrics Bulletin*, *2*(6).

[13] Wald, A. (1943). Tests of statistical hypotheses concerning several parameters when the number of observations is large. *Trans. AMS*, *54*(3).

[14] Welch, B. L. (1947). The generalization of “Student’s” problem when several different population variances are involved. *Biometrika*, *34*(1–2).